Team Number: 135

M1: Approach — Green Customer Insights Platform

1. Problem and our understanding

Build an AI platform that finds customers who could benefit from green financial products (EV Finance, Green Home Loan, Solar Finance, Paperless Banking), estimates the sustainability impact, and explains every recommendation to customers and business teams.

We treat the core rule as a strict split of roles:

- **ML models** predict: transaction category and take-up propensity.
- **A rules-based carbon engine** calculates all impact using documented formulas and emissions factors.
- **The LLM** only explains structured outputs. It never predicts, calculates or decides eligibility.

Propensity is not eligibility, and every impact figure is an estimate.

2. Idea

Every recommendation is one structured record: *customer, product, propensity, confidence, reason codes, evidence, estimated impact, data timestamp*. A rules layer first removes products the data rules out (e.g. EV finance for an existing EV owner). The LLM then turns the record into plain language, and its output is checked so every number and reason matches the record. If data is missing, the system says “insufficient information” instead of guessing.

3. Approach and pipeline

Synthetic + public data	**Customer 360**	**Transaction categoriser**	**Propensity models**
Carbon engine	**Rules + recommender**	**LLM explanation**	**API + dashboard**

- **Data:** synthetic customer, transaction, product, vehicle, energy and digital-usage tables, shaped using public datasets (London Smart Meter, UCI Online Retail II). No real banking data. Take-up labels follow documented rules with added noise, within a 6-month outcome window after a fixed cutoff date.
- **Transaction categorisation:** merchant-keyword rules plus a text classifier, giving exactly one of the 8 categories, with written boundary rules (e.g. Fuel vs Automotive).
- **Propensity:** one gradient-boosted model per product, calibrated. Confidence combines calibration reliability, model agreement and data completeness. Reason codes come from the top contributing features.
- **Carbon engine:** documented formulas using UK Government (DEFRA) emissions factors, following GHG Protocol avoided-emissions guidance. Emissions, energy, paper and digital adoption are reported separately.

- **Outputs:** a customer view (insight, evidence, impact, action) and a business dashboard (segments, opportunities, impact) with drill-down from customer to action.

4. Testing methodology

What	How	Metrics
Data	Customer-level train/validation/test split (70/15/15); features only from before the cutoff to avoid leakage.	Leakage checks, plausibility checks
Categorisation	Held-out noisy merchant text, including boundary cases.	Macro-F1, confusion matrix
Propensity	Compare against a simple baseline on validation data; final check on the test set.	ROC-AUC, PR-AUC, Brier, lift
Carbon engine	Unit tests against hand-calculated cases.	Exact match, units and sources shown
LLM	Fixed evaluation set with automatic checks against the input record.	Groundedness, product mapping, completeness
Scenarios	Missing data, conflicting signals, ruled-out products, insufficient data for impact.	Correct handling, reproducible rerun

Stack: Python, Pandas, scikit-learn, LightGBM, SQLite, FastAPI, Streamlit, Plotly, one hosted LLM API.